

Srinjoy Chakraborty

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Portfolio

Education

Sant Longowal Institute of Engineering and Technology (SLIET)

B.E. in Electrical Engineering
Sangrur, Punjab

GPA: 7.79/10
Nov 2021 - Present

Experience

Undergraduate Research Project Associate

Jan 2025 - Present

PI: Dr. R.K. Mittal

Multi-scale Manufacturing Lab, Indian Institute of Technology, Guwahati
Guwahati, India

- Creating an automated focus-stacking and inspection system for micro-machined surfaces and precision tools by integrating high-precision mechatronics, real-time feedback control, and custom software—resulting in precise 3D reconstructions and scalable measurements.
- Refining focus-stacking algorithms and implementing precise control mechanisms, significantly boosting the accuracy and efficiency of 3D point cloud generation.

Summer Research Intern

Jun 2024 - Dec 2024

PI: Dr. R.K. Mittal

Multi-scale Manufacturing Lab, Indian Institute of Technology, Guwahati
Guwahati, India

- Converted 2D images into scalable 3D point clouds, achieving a 92% scaling accuracy.
- Engineered an image processing pipeline to isolate in-focus regions, thereby enhancing the quality of 3D reconstructions.
- Deployed Mask R-CNN and Laplacian edge detection algorithms to generate accurate masks and execute focus stacking, which improved composite 3D image quality.
- Leveraged OpenCV for optimizing image alignment and fusion—employing brute-force matching and SIFT techniques to streamline 3D point cloud creation.

Summer Research Intern

Jun 2023 - Jul 2023

PI: Dr. Nilanjan Chattaraj

Department of Electronics and Communication Engineering, National Institute of Technology, Durgapur, India

- Designed and implemented ROS nodes and scripts to enable autonomous navigation for a differential drive robot using SLAM.
- Utilized ROS, SLAM, Gazebo, Python, C++ to develop and test the system.

Winter Intern

Jan 2023 - Feb 2023

Indian Railways, Alipurduar Junction, West Bengal

- Executed maintenance and operational protocols for transformers, synchronization panels, water pumps, and diesel generators.
- Diagnosed and resolved technical issues in synchronization panels and DG sets, reducing system downtime.

Publications

- **DSDMA: Dynamic Spray Drift Mitigation Algorithm for Optimizing Spraying Drone Performance in Dynamic Agricultural Environments**

Transactions on Intelligent Systems and Technology (Under Review)

Authors: Goyal, R.; **Chakraborty, S.**; Nath, A.; Niranjana, U.

Keywords: Precision Agriculture, UAV Trajectory Optimization, AirSim Simulation

Summary: Co-authored a research manuscript that introduces a novel algorithm to optimize spraying drone performance by dynamically mitigating spray drift in agricultural settings.

Projects

SLAMDrive UAV, IoRT and AI Lab, SLIET

May 2024 - Jan 2025

PI: Dr. Amar Nath, Dept of CSE, SLIET

Tech Stack: ROS2, Python, PyTorch, OpenCV, R-CNN, AirSim

GitHub

- Engineered an autonomous UAV leveraging RetinaNET for weed detection in potato farms (89% accuracy) and human surveillance (97% accuracy) using real-time image processing and mapping.
- Devised a custom aerodynamic compensation algorithm via the AirSim Python API, optimizing fertilizer dispersal in windy conditions and achieving 97% path traversal rectification accuracy.
- Simulated complex operational environments using AirSim and ROS2 to validate real-time performance.
- Integrated Dijkstra's algorithm and the Bendy Ruler technique for efficient path planning, incorporated Hector SLAM for dynamic mapping and obstacle avoidance and programmed automated takeoff, landing, and Return-to-Home (RTH) functions.

LSTM-RNN based Emotion Detection System

Oct 2024 - Dec 2025

PI: Dr. Surita Maini, Dept of EIE, SLIET

Tech Stack: Python, PyTorch, LSTM

GitHub

- Engineered a comprehensive preprocessing pipeline using ICA for artifact removal, bandpass/notch filtering, and min-max normalization, significantly enhancing EEG signal quality.
- Constructed and optimized an LSTM-RNN model for real-time emotion recognition, achieving 90.9% test accuracy on the SEED-IV dataset by leveraging temporal EEG dependencies and fine-tuning hyperparameters with TensorFlow/Keras.

Holonomic Drive Bot

Nov 2023 - Feb 2024

Tech Stack: ROS2, Gazebo, OpenCV, Easy EDA

GitHub

- Engineered and coordinated a fleet of three holonomic drive robots to autonomously render large glyphs on an 8 ft x 8 ft arena, employing overhead camera localization and Aruco markers for precise navigation.

Self-Balancing Bot

Nov 2022 - Feb 2023

Tech Stack: CoppeliaSim, Lua, Octave

GitHub

- Developed a self-balancing bot utilizing the LQR algorithm, enabling robust traversal of steep slopes and sharp turns.

Technical Skills

- **Programming Languages:** Python, C/C++, Lua, SQL, Arduino IDE
- **Machine Learning & Computer Vision:** Scikit-learn, Keras, OpenCV, TensorFlow
- **Engineering Tools:** MATLAB, Simulink, Octave, Gazebo, CoppeliaSim, ArduPilot, Mission Planner

Achievements and Certifications

- **Certifications:** MATLAB Simulink, MATLAB Stateflow, Machine Learning (Deeplearning.ai)
- **Achievements:** Contributed to NASA Asteroid Search Campaign with 11 preliminary discoveries; earned accolades at technical fests and robotics competitions (SLIET, IIT BHU, IIT Delhi)